

RAHAGEER SAADMAN ISLAM

AI / ML Engineer | Agentic AI · LangChain · LLMs · RAG · MCP · MLOps · FastAPI · Python

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PROFESSIONAL SUMMARY

AI/ML Engineer with hands-on production experience building autonomous AI agent systems, Retrieval-Augmented Generation (RAG) pipelines, and scalable MLOps infrastructure. Currently deploying enterprise-grade AI capabilities at Deep Data Labs (Cloudscockpit) using LangChain, LangGraph, and Model Context Protocol (MCP) — with 6+ MCP tools shipped to production. Deep expertise in large language models (LLMs), neural search, vector embeddings, NLP, and computer vision, backed by end-to-end project delivery from model development to REST API deployment on AWS. Seeking an AI/ML Engineer role to build intelligent, production-ready systems at scale.

TECHNICAL SKILLS

AI / ML Core:	Large Language Models (LLMs), NLP, Generative AI, Computer Vision, Explainable AI (XAI), Anomaly Detection, Classification
Agentic AI & RAG:	LangChain, LangGraph, Model Context Protocol (MCP), ReAct Agents, RAG, Vector Embeddings, Semantic Search
ML Libraries:	Scikit-learn, Pandas, NumPy, Jupyter Notebook — feature engineering, cross-validation, ensemble methods
APIs & LLM Providers:	Google Gemini API, Serper API, Tavily API, NewsAPI, OpenAI-compatible endpoints, FastAPI, REST API Design
MLOps & Cloud:	Python (98.5%), Docker, Kubernetes, AWS (ECS, S3, ElastiCache, RDS, Lambda), Redis, PostgreSQL, CI/CD for ML

WORK EXPERIENCE

AI/ML Engineer (MLOps) — Deep Data Labs / Cloudscockpit

Dhaka, Bangladesh | January 2026 – Present

- Designed and deployed 6+ production MCP-based AI tools (neural search, news aggregation, claim verification, computer vision) into the Cloudscockpit genai-core platform — PR #127 merged to main.
- Built semantic RAG pipelines using LangChain and vector embeddings with tier-wise embeddings support, enabling context-aware retrieval for enterprise AI agents.
- Developed scalable FastAPI microservices for AI agent orchestration with Redis caching, PostgreSQL connection pooling, and AWS ECS deployment for high-availability production environments.
- Integrated Google Gemini API for multi-modal image analysis, extending the Aura MCP agent framework with computer vision capabilities.
- Implemented LangGraph-based stateful agent workflows for complex multi-step AI reasoning and autonomous task execution.
- Contributed to MLOps infrastructure optimization: fixed critical MCP tool injection bugs, enhanced knowledge-base scaling, and improved system reliability for production AI pipelines.

Stack: Python, FastAPI, LangChain, LangGraph, MCP, Vector Databases, Redis, PostgreSQL, AWS ECS/S3/ElastiCache, Google Gemini API

AI / ML PROJECTS

Fact-Check Agent — Autonomous Multi-Source Verification System

github.com/Rahageer90/fact_check_agent | Python · FastAPI · LangChain · MCP · Google Gemini · React

- Architected a full-stack agentic AI system for autonomous claim verification, orchestrating multiple LLM providers (Google Gemini API) with real-time data retrieval.
- Engineered 6+ MCP-based tools: neural search, news aggregation (NewsAPI), web search (Serper API), and structured claim validation — each with fallback mechanisms for production reliability.
- Built ReAct agent loop with LangChain for multi-step reasoning: query decomposition, evidence retrieval, cross-source reconciliation, and confidence-scored verdict generation.

- Deployed FastAPI REST backend and React frontend; demonstrates enterprise-scale agentic system architecture with extensible, modular MCP tooling.

Stack: Python, FastAPI, LangChain, MCP, Google Gemini API, Serper API, NewsAPI, React, REST API

Explainable AI (XAI) — Integrated Network Intrusion Detection System

github.com/Rahageer90/ML_EXAi_-Integrated_IDS_Project | Python · Scikit-learn · XAI · Jupyter Notebook

- Developed ensemble machine learning models (classification + anomaly detection) to identify malicious network traffic patterns across multiple attack categories.
- Implemented Explainable AI (XAI) techniques — feature importance analysis and model interpretability tools — ensuring transparent decision-making critical for security applications.
- Delivered complete ML pipeline: data preprocessing, feature engineering, model training, cross-validation, hyperparameter tuning, and performance benchmarking.

Stack: Python, Scikit-learn, XAI / SHAP, Pandas, NumPy, Jupyter Notebook

Healthcare ML — Diabetes Risk Prediction System

github.com/Rahageer90/Diabetes_Prediction | Python · Scikit-learn · Pandas · Jupyter Notebook

- Engineered a supervised ML model for diabetes risk assessment: full pipeline covering data cleaning, feature engineering, model selection, and statistical validation.
- Applied cross-validation and hyperparameter tuning to maximize generalizability and reliability of predictions on clinical datasets.

Stack: Python, Scikit-learn, Pandas, NumPy, Jupyter Notebook

EDUCATION

B.Sc. in Computer Science & Engineering — BRAC University

Dhaka, Bangladesh

Oct 2021 – Jan 2026

Higher Secondary Certificate, Science — Adamjee Cantonment College

Dhaka, Bangladesh

Jul 2018 – Apr 2020

CERTIFICATIONS

- Python for Data Science, AI & Development — IBM / Coursera
- Data Analysis with Python — IBM / Coursera
- Introduction to Generative AI — Google / Coursera
- Applied Data Science Capstone — Coursera